MATH / BC GRADE 11

Quadratic functions

Find roots and a vertex, and connect them to a parabola.

Pre-calculus 11

01 Recognize a quadratic

A quadratic function has the form $y = ax^2 + bx + c$, with a not equal to zero. Its graph is a parabola. Positive a opens upward; negative a opens downward.

02 Use vertex form

In $y = a(x - h)^2 + k$, the vertex is (h,k), and the axis of symmetry is $x = h$. The vertex is a minimum for $a > 0$ and a maximum for $a < 0$.

03 Solve by factoring

To find x-intercepts, set $y = 0$. If a product is zero, at least one factor is zero. This zero-product rule applies after the equation is set equal to zero.

Worked example

Explore $y = x^2 - 6x + 8$

Factor: $y = (x - 2)(x - 4)$. The roots are 2 and 4. Complete the square: $y = (x - 3)^2 - 1$. The vertex is (3,-1); the axis is $x = 3$. The graph opens upward.

Watch out: In $(x - 3)^2$, the vertex x-coordinate is +3, not -3.

MATH / BC GRADE 11

Try it. Then check it.

Work through these questions before looking at the answers.

1. Find the roots of $x^2 - 5x + 6 = 0$.

2. State the vertex and opening direction of $y = -2(x + 1)^2 + 4$.

3. Does $y = (x - 2)^2 + 3$ have any real x-intercepts? Explain.

Answer key

1. Factor: $(x - 2)(x - 3) = 0$. Thus $x = 2$ or $x = 3$.

2. Vertex (-1,4); opens downward because $a = -2$.

3. No. The square is at least zero, so y is at least 3 and cannot equal zero.

One-minute recap

Close the notes. Explain one key idea in your own words, then retry the question you found most difficult.

About this guide

Original Study Bridge topic review prepared with AI assistance. Suggested course fit, not complete or official course coverage. Compare with your teacher's materials. Examples and practice questions are newly written.

Official course context: Pre-calculus 11

Questions or corrections: ivyystudyy@gmail.com